

SECTION A: Answer ALL Questions (40 Marks)

1. Explain the filter-feeding mechanism of sessile invertebrates living in stagnant water.

Sessile invertebrates, such as sponges and barnacles, use specialized structures to filter particles from water for nutrition. Sponges, for instance, use choanocytes (collar cells) to create a current that pulls water through pores into their central cavity. Microorganisms and organic particles are trapped and ingested, while clean water is expelled.

2. State five benefits of annelids to humans.

1. **Soil aeration:** Earthworms improve soil structure and fertility.
 2. **Decomposition:** They break down organic matter, enriching soil nutrients.
 3. **Fishing bait:** Many species are used as bait for fishing.
 4. **Medical use:** Leech saliva contains anticoagulants used in medicine.
 5. **Scientific research:** Annelids are studied for understanding regeneration and biology.
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3. Explain five protective mechanisms associated with arthropods.

1. **Exoskeleton:** Hard outer covering protects from predators and dehydration.
 2. **Camouflage:** Blending with surroundings to avoid detection (e.g., stick insects).
 3. **Venom:** Many arthropods (e.g., spiders, scorpions) use venom to deter predators.
 4. **Flight:** Insects like butterflies and bees escape predators by flying.
 5. **Mimicry:** Some arthropods imitate harmful species to avoid predation.
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4. Describe radial symmetry.

Radial symmetry is a body plan where parts are arranged around a central axis, like spokes on a wheel. Organisms with this symmetry (e.g., jellyfish, starfish) can interact with their environment from all directions.

5. Explain why marine water supports more invertebrate life than freshwater.

- Marine water has stable temperatures and abundant minerals.
- It provides a consistent supply of dissolved salts essential for osmoregulation.

- High biodiversity in ecosystems like coral reefs offers niches for various invertebrates.
 - Marine environments are vast, supporting larger populations and diversity.
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6. (a) Describe the process of cephalization in invertebrates.

Cephalization is the evolutionary trend of concentrating sensory organs and nerve cells at the anterior (head) end. This process is seen in flatworms and arthropods, enhancing their ability to detect and respond to stimuli efficiently.

(b) State the role it plays.

Cephalization improves sensory perception, movement coordination, and predator-prey interactions.

7. Draw a well-labeled diagram of a named ciliate protozoan.

Example: **Paramecium** (include key features such as cilia, pellicle, contractile vacuole, oral groove, and macronucleus).

8. (a) Describe the formation of a pseudocoelom.

A pseudocoelom forms when the body cavity is only partially lined with mesoderm. It is found in roundworms (Nematoda) and provides space for organ development, distribution of nutrients, and movement.

(b) State the role it plays.

1. Acts as a hydrostatic skeleton, aiding movement.
 2. Provides space for organ growth.
 3. Facilitates nutrient and waste transport.
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SECTION B: Answer any ONE Question (30 Marks)

9. Write an essay on the adaptive features and classification of echinoderms.

Adaptive Features of Echinoderms:

1. **Water vascular system:** Used for locomotion, feeding, and gas exchange.
2. **Tube feet:** Aid in movement and attachment to surfaces.
3. **Spiny skin:** Offers protection from predators.

4. **Regeneration:** Ability to regrow lost parts (e.g., arms of starfish).
5. **Radial symmetry:** Helps them interact equally in all directions.
6. **Internal skeleton:** Made of calcium carbonate for support.
7. **Pentaradial structure:** Allows flexibility and functionality.
8. **Low metabolic rate:** Helps them survive in nutrient-scarce environments.
9. **Filter-feeding:** Some species like sea cucumbers filter nutrients from water.
10. **Ability to adapt to depth:** Found at various ocean depths.

Classification:

1. **Class Asteroidea:** Starfish.
 2. **Class Echinoidea:** Sea urchins and sand dollars.
 3. **Class Holothuroidea:** Sea cucumbers.
 4. **Class Crinoidea:** Feather stars.
 5. **Class Ophiuroidea:** Brittle stars.
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10. Discuss the adaptive features of protozoans.

1. **Locomotion structures:** Use cilia, flagella, or pseudopodia for movement.
 2. **Protective cysts:** Form cysts to survive unfavorable conditions.
 3. **Efficient reproduction:** Reproduce quickly through binary fission.
 4. **Flexible nutrition:** Heterotrophic (e.g., amoeba) or autotrophic (e.g., euglena).
 5. **Contractile vacuoles:** Regulate water balance to prevent bursting.
 6. **Small size:** Easily adapt to microscopic environments.
 7. **Symbiosis:** Some live in mutualistic relationships (e.g., termite gut protozoans).
 8. **Parasitism:** Adapted to live within hosts (e.g., Plasmodium in humans).
 9. **Rapid adaptability:** Evolve quickly to changing environments.
 10. **Specialized organelles:** Structures like pellicles for protection.
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11. Marine invertebrates float on the surface of water or live at the bottom. Discuss this statement.

1. **Surface dwellers:**
 - Adaptations include flattened bodies to increase buoyancy.
 - Air-filled sacs (e.g., Portuguese man-o-war).
 - Plankton-like behavior to drift with currents.
2. **Bottom dwellers:**
 - Adapted with sturdy shells for protection.
 - Burrowing ability (e.g., clams).
 - Filter-feeding mechanisms to consume detritus.
3. **Middle-dwellers:**
 - Swim actively (e.g., jellyfish).

Environmental conditions like pressure, light, and food sources influence distribution.

SECTION A: Answer ALL Questions (40 marks)

1. State five unique characteristics of asteroid echinoderms.

1. **Pentamerous symmetry:** Their body is organized in five or multiples of five arms.
 2. **Water vascular system:** Used for movement, feeding, and respiration.
 3. **Tube feet:** Located on the underside of arms, aiding in locomotion and food handling.
 4. **Regenerative ability:** Can regrow lost arms.
 5. **Endoskeleton:** Made of calcareous plates, providing structure and protection.
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2. Describe the various types of symmetry in invertebrates.

1. **Radial symmetry:** Body parts arranged around a central axis (e.g., jellyfish).
 2. **Bilateral symmetry:** Divided into two equal halves along one plane (e.g., insects).
 3. **Asymmetry:** No specific symmetry (e.g., sponges).
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3. Explain the hydrozoan life cycle.

Hydrozoans, such as *Obelia*, exhibit alternation of generations:

- **Polyp stage:** Sessile and asexual, producing medusae by budding.
 - **Medusa stage:** Free-swimming and sexual, releasing gametes for fertilization.
 - **Planula larva:** Resulting from fertilization, which settles and develops into a new polyp colony.
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4. List five major differences between protostomes and deuterostomes.

1. **Embryonic development:** Protostomes develop the mouth first; deuterostomes develop the anus first.
 2. **Cleavage:** Protostomes show spiral cleavage; deuterostomes have radial cleavage.
 3. **Coelom formation:** Protostomes form coelom by splitting; deuterostomes by outpocketing.
 4. **Nervous system:** Protostomes often have a ventral nerve cord; deuterostomes have a dorsal nerve cord.
 5. **Examples:** Protostomes include arthropods and mollusks; deuterostomes include echinoderms and chordates.
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5. Outline five importance of protozoans to humans.

1. **Nutrient cycling:** Decompose organic matter, enhancing nutrient availability.
 2. **Food chain:** Serve as food for aquatic organisms like fish.
 3. **Medical research:** Model organisms for studying diseases.
 4. **Pathogens:** Some (e.g., *Plasmodium*) help in understanding human diseases like malaria.
 5. **Symbiotic relationships:** Aid digestion in animals like termites.
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6. Describe briefly the three categories of triploblastic animals based on their internal body cavity.

1. **Acoelomates:** No body cavity; organs embedded in solid mesoderm tissue (e.g., flatworms).
 2. **Pseudocoelomates:** Have a body cavity not fully lined by mesoderm (e.g., roundworms).
 3. **Coelomates:** True body cavity completely lined by mesoderm (e.g., annelids).
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7. Explain the adaptations of spiders to their predatory way of life.

1. **Silk production:** Spiders spin webs to trap prey.
 2. **Venom:** Inject venom to paralyze or kill prey.
 3. **Keen eyesight:** Detects movement of prey.
 4. **Leg structures:** Designed for swift movements and capturing prey.
 5. **Camouflage:** Blend with surroundings to ambush prey.
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8. State five characteristics of sponges that show their evolutionary primitiveness.

1. **Lack of tissues and organs:** Rely on specialized cells for functions.
 2. **Sessile nature:** Permanently attached to substrates.
 3. **Asymmetry:** Some exhibit no symmetry.
 4. **Filter feeding:** Obtain nutrients by filtering water through pores.
 5. **Reproduction:** Can reproduce sexually and asexually.
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SECTION B: Answer any ONE Question (30 marks)

9. Discuss the adaptive features of Onychophorans.

1. **Soft, flexible body:** Adapted for movement in leaf litter and crevices.
2. **Hydrostatic skeleton:** Supports movement.
3. **Cuticle:** Protects against desiccation.
4. **Leg-like appendages:** Assist in crawling.
5. **Oral papillae:** Used to eject slime for capturing prey or defense.

6. **Nocturnal behavior:** Reduces predation risk.
 7. **Respiratory system:** Tracheae allow efficient gas exchange.
 8. **Moist habitats:** Prevent water loss.
 9. **Slime production:** Unique feature for predation and defense.
 10. **Carnivorous diet:** Adapted to capture and consume small invertebrates.
 11. **Segmented body:** Offers flexibility.
 12. **Sensory antennae:** Detect prey and environmental changes.
 13. **Simple reproductive system:** Ensures survival of offspring.
 14. **Low metabolic rate:** Suitable for nutrient-scarce environments.
 15. **Camouflage:** Blends with surroundings for protection.
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10. Write an essay on Oligochaetes.

Introduction:

Oligochaetes, like earthworms, are segmented worms belonging to the phylum Annelida. They are primarily terrestrial but can also be aquatic.

Characteristics and Adaptive Features:

1. **Segmented body:** Enhances flexibility and movement.
 2. **Setae:** Small bristles help in anchoring and locomotion.
 3. **Moist skin:** Essential for gas exchange.
 4. **Hermaphroditic reproduction:** Increases reproductive success.
 5. **Burrowing behavior:** Aids in aerating the soil.
 6. **Digestive system:** Specialized for consuming and decomposing organic matter.
 7. **Closed circulatory system:** Efficient nutrient and oxygen transport.
 8. **Longitudinal and circular muscles:** Aid in peristaltic movement.
 9. **Adaptation to nocturnal behavior:** Reduces predation risk.
 10. **Simple nervous system:** Coordinates movement and response to stimuli.
 11. **Resistance to harsh conditions:** Form cocoons during unfavorable seasons.
 12. **Habitat versatility:** Thrive in various soil types.
 13. **Soil fertilization:** Enhance nutrient availability.
 14. **Low energy requirements:** Suited for nutrient-poor environments.
 15. **Symbiosis with microorganisms:** Aid in digestion and soil health.
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11. Slime molds are classified as both animals and fungi. Discuss this statement.

1. **Fungal-like features:**
 - Reproduce using spores.
 - Thrive in moist environments.
 - Feed on decaying organic matter.
2. **Animal-like features:**
 - Motile during their plasmodial stage.
 - Engulf food particles via phagocytosis.
3. **Lifecycle stages:**

- Exhibit a mix of fungal and animal traits, transitioning between stationary and mobile forms.
- 4. **Single-cell and colony form:** Behave like unicellular organisms but can aggregate to form multicellular structures.
- 5. **Decomposers:** Contribute to nutrient cycling like fungi.
- 6. **Pseudopodia:** Extend projections for movement and feeding.
- 7. **Lack of cell walls:** Differentiates them from true fungi.
- 8. **Amoeboid movement:** Similar to protists or primitive animals.
- 9. **Multinucleated stage:** Characteristic of plasmodial slime molds.
- 10. **Moist habitat adaptation:** Common to both fungi and certain animal-like protists.
- 11. **Sensitivity to environmental signals:** Aggregation occurs under stress.
- 12. **Reproductive cycle:** Includes asexual spore production and sexual stages.
- 13. **Cytoplasmic streaming:** Aids in nutrient and oxygen distribution.
- 14. **Evolutionary link:** Represent an intermediate between fungi and animals.
- 15. **Biological significance:** Provide insights into evolution, cell signaling, and cooperation.